
LLMWiki DeepEval Benchmark: Updated Final Report

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Abstract

This updated report summarizes the final benchmark configuration of the LLMWiki DeepEval project on TriviaQA RC (250 samples). Earlier exploratory runs were removed from the main report because they no longer reflect the current metric implementation. The final pipeline uses a hybrid Answer Relevancy metric (local LLM judge + embedding-like similarity + lexical signal), alongside Faithfulness, Exact Match (EM), F1, Time-To-First-Token (TTFT), and Tokens-Per-Second (TPS). Final results show strong grounding and solid answer accuracy: Answer Relevancy 0.6190, Faithfulness 0.9461, EM 0.7160, and F1 0.7969.

Attestation of Teamwork

Axel Tang did all this and learnt with AI

1 Executive Summary

This report now focuses only on the latest benchmark artifact:

- **Run timestamp:** 20260513_023249
- **Dataset:** TriviaQA RC
- **Samples evaluated:** 250
- **Model:** Qwen 3.5-35B through LMStudio

Earlier runs were intentionally removed from this document because they used older metric behavior and are not representative of the current evaluation setup.

2 Methodology

2.1 Task and Data

The benchmark evaluates question answering on TriviaQA RC using provided search evidence. For each question, the model receives context and generates a concise answer.

2.2 Metrics

The final report uses six metrics:

FujiFilm (Axel)

- **Answer Relevancy (Hybrid)**: weighted combination of
 - local LLM judge relevance score,
 - embedding-like token-frequency cosine similarity,
 - lexical overlap signal.
- **Faithfulness**: answer-context grounding score.
- **Exact Match (EM)**: strict answer match against aliases.
- **F1**: token-level overlap with reference answers.
- **TTFT**: latency to first token.
- **TPS**: generation throughput.

The Answer Relevancy metric uses dynamic weighting so missing components do not invalidate scoring.

2.3 Run Configuration

Table 1: Final Benchmark Configuration

Parameter	Value
Model	qwen/qwen3.5-35b-a3b
Host	10.86.30.140:1234
Dataset	TriviaQA RC
Sample count	250
Context source	ranked TriviaQA search results

3 Final Results

3.1 Aggregate Scores

Table 2: Final Run Metrics (20260513_023249)

Metric	Value
Answer Relevancy (Hybrid)	0.6190
Faithfulness	0.9461
Exact Match (EM)	0.7160 (71.60%)
F1	0.7969
TTFT	259.60 ms
TPS	5.28

3.2 Answer Relevancy Components

Table 3: Answer Relevancy Component Means

Component	Mean
Local LLM Judge	0.9248
Embedding-like Similarity	0.0475
Lexical Signal	0.0601
Hybrid Output	0.6190

Interpretation. The hybrid score resolves the previous under-reporting issue seen in earlier lexical-heavy scoring. Final relevance (0.6190) is now directionally consistent with EM/F1 and substantially more informative than previous near-zero values.

4 Discussion

4.1 What the Final Numbers Mean

- **High faithfulness (0.9461)** indicates strong grounding in retrieved context.
- **EM/F1 (0.7160/0.7969)** indicate solid extractive QA performance with alias matching.
- **Relevancy (0.6190)** indicates moderate-to-good question-answer alignment under the updated semantic-aware metric.

4.2 Performance Notes

TTFT and TPS indicate acceptable interactive behavior for this setup, though throughput can still be improved through model serving and batching optimizations.

4.3 How Metric Combinations Affect RAG Quality

Single metrics are useful, but combinations are more diagnostic for RAG systems.

4.3.1 Relevance-Faithfulness Matrix

Table 4: Interpretation of Relevance and Faithfulness Combinations

Answer Relevancy	Faithfulness	Interpretation for RAG Quality
High	High	Best case. Retrieval and generation are aligned; answers are both on-topic and grounded.
High	Low	Fluent but risky. Model answers the question style-wise, but may hallucinate unsupported details.
Low	High	Safe but unhelpful. Output is grounded in retrieved text, but may miss the user intent.
Low	Low	Failure mode. Retrieval and generation are both weak for the task.

4.3.2 ELI5: Relevance-Faithfulness Matrix

- **High Relevance + High Faithfulness:** The system found the right page and copied the right fact. This is what we want.
- **High Relevance + Low Faithfulness:** It sounds like a good answer to your question, but it may be making things up.
- **Low Relevance + High Faithfulness:** It is telling true facts from the documents, but not the fact you asked for.
- **Low Relevance + Low Faithfulness:** It is both off-topic and unreliable; this is a bad answer.

4.3.3 EM/F1 Regimes

- **High EM, High F1:** Strong extractive correctness and robust token overlap; ideal for factoid QA.
- **Low EM, High F1:** Near-miss regime; answer is often close but differs by formatting, aliasing, or small wording choices.
- **Low EM, Low F1:** Core correctness issue; retrieval quality, prompting, or answer extraction likely needs work.
- **High EM, Lower Relevancy:** Usually indicates metric mismatch or calibration issues in relevancy scoring rather than catastrophic model behavior.

4.3.4 ELI5: EM/F1 Regimes

- **High EM, High F1:** The answer is basically exactly right.
- **Low EM, High F1:** The answer is close, but wording is a little off (like spelling or alias differences).
- **Low EM, Low F1:** The answer is mostly wrong.
- **High EM, Lower Relevancy:** The answer might still be correct, but the relevancy scoring method may need tuning.

4.3.5 Latency/Throughput Effects on User Experience

- **Lower TTFT** improves perceived responsiveness.
- **Higher TPS** improves completion speed and throughput under load.
- A system can be accurate but still feel poor in production if TTFT is high or TPS is low.

4.3.6 ELI5: TTFT and TPS

- **TTFT** is how long you wait before seeing the first word.
- **TPS** is how fast the rest of the words come out.
- Even a smart system feels bad if it starts too slowly or types too slowly.

5 Conclusions

The final LLMWiki DeepEval pipeline demonstrates:

1. strong grounding quality,
2. solid extractive QA accuracy,
3. improved and more trustworthy answer relevance scoring after metric updates.

The updated metric behavior supports using this pipeline for continued benchmarking and iterative RAG optimization.

Artifacts

- Final benchmark JSON: `benchmark_results/llmwiki_deepeval_benchmark_20260513_023249.json`
- Final simplified JSON: `benchmark_results/llmwiki_deepeval_simplified_20260513_023249.json`
- Final CSV: `benchmark_results/llmwiki_deepeval_results_20260513_023249.csv`
- Notebook: `llmwiki_deepeval_benchmark.ipynb`

Reproducibility Commands

```
cd /Users/axelmacbookair/Desktop/LLMWiki_DeepEval
source .venv/bin/activate
jupyter notebook
# Open llmwiki_deepeval_benchmark.ipynb and run all cells
```

Appendix: Metric Equations

Exact Match (EM)

$$\text{EM} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\hat{a}_i \in \mathcal{A}_i\}$$

where $\hat{a}_i$ is the prediction and $\mathcal{A}_i$ is the alias set for sample i .

F1 Score (Token Level)

For each sample:

$$\begin{aligned} \text{Precision} &= \frac{|T(\hat{a}) \cap T(a)|}{|T(\hat{a})|}, \quad \text{Recall} = \frac{|T(\hat{a}) \cap T(a)|}{|T(a)|} \\ \text{F1} &= \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned}$$

Then report mean F1 across samples (or max-over-aliases per sample before averaging).

Answer Relevancy (Hybrid)

$$\text{Rel}_{\text{hybrid}} = \sum_{k \in K} \hat{w}_k s_k, \quad K \subseteq \{\text{llm}, \text{embed}, \text{lex}\}, \quad \hat{w}_k = \frac{w_k}{\sum_{j \in K} w_j}$$

Unavailable components are dropped and weights are re-normalized over available components.

Faithfulness

Using token-overlap grounding (as implemented):

$$\text{Faithfulness} = \frac{|T(\hat{a}) \cap T(c)|}{|T(\hat{a})|}$$

where $T(\hat{a})$ are answer tokens and $T(c)$ are context tokens after normalization.

Time To First Token (TTFT)

$$\text{TTFT}_i = t_{\text{first-token},i} - t_{\text{request-start},i}$$

Aggregate TTFT is typically the mean over samples.

Tokens Per Second (TPS)

$$\text{TPS}_i = \frac{\text{completion_tokens}_i}{\text{total_generation_time}_i}$$

Aggregate TPS is typically the mean over samples.